

Bonnabelle Zabelle

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—Profile—

A skilled data scientist and researcher with 3+ years of experience in applied statistics, through the lens of astrophysics research. I am hoping to apply my experience with large, complex data sets and expand my knowledge with more earthly applications.

—Experience—

Research Assistant, Python Developer

Minnesota Institute for Astrophysics, Twin Cities

Minneapolis, MN

January 2021 – January 2025

- Developed code infrastructure to allow basic users to conduct complex queries on an unstructured database.
- Ran large scale simulations of observations from the Euclid Space telescope to create a user-friendly table to quickly estimate future observations.
- Collaborated with a team of statisticians to create a novel Bayesian and Frequentist approach to distance measurement in Astrophysics. Served as the astrophysics reference and lead python developer.
- Planned and saw through a research project at Keck Observatory, including the resulting data reduction and analysis.

Bayesian Astrostatistics Teaching Assistant

University of Minnesota, Twin Cities

Minneapolis, MN

August 2022 – December 2023

- Taught the foundations of Bayesian statistics through the lens of Astrophysics. Focused on modern techniques of MCMC, bayesian model selection, and hierarchical modeling.
- Worked with Dr. Scarlata to create the course textbook, including testing and coding examples in R and Python.

Administrative Assistant

University of Minnesota, Twin Cities

Minneapolis, MN

May 2021 – July 2022

- Administrator for the UMN Physics Force and Physics Careers, managing a range of tasks from setting up shows, career and outreach events, as well as maintaining the websites and emails.
- Created a comparative report, including data acquisition and analysis, comparing the UMN undergraduate physics program to other American institutions.

—Skills—

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|-----------|-----------|----------------------|
| • Python | • Jupyter | • Scientific Writing |
| • R | • rStudio | • Communication |
| • SQL | • Git | • Public Speaking |
| • PyTorch | • LaTeX | • Time Management |

—Publications—

- Bonnabelle Zabelle *et al* 2023 *ApJ* 947 17
- Vihang Mehta *et al* 2024 *ApJS* 275 17
- Lei Sun *et al* 2024 *ApJ* 972 8

—Education—

University of Minnesota - College of Science and Engineering

Master of Science - Astrophysics; GPA 3.772

Awarded Data Science and Multi-messenger Astronomy Fellowship

Minneapolis, Minnesota

Received January 2025

University of Minnesota - College of Science and Engineering

Bachelor of Science - Astrophysics; GPA 3.209

Minneapolis, Minnesota

Received August 2021